



**DEPARTMENT OF THE AIR FORCE
AIR FORCE LIFE CYCLE MANAGEMENT CENTER
WRIGHT-PATTERSON AIR FORCE BASE**

March 12, 2026

Tracking No. RFI FA8629-26-R-5007

MEMORANDUM FOR: All Qualified Vendors with SAM.gov Access

FROM: AFLCMC/WISK
Special Warfare Contracting Office
1895 Fifth Street
Wright Patterson AFB, OH 45433-7200

SUBJECT: Request for Information (RFI) – Multiple in Multiple Out (MIMO) signal Mobil Ad Hoc Network (MANET)

1. NOTICE

This Sources Sought Synopsis (SSS)/Request for Information (RFI) is issued for informational purposes and market research only; it does not constitute a solicitation. The Government will not reimburse any company or individual for any expenses associated with preparing/submitting information in response to this posting. The information provided may be used by the Government in developing its acquisition strategy, statement of work/statement of objectives, and/or performance specifications. Interested parties are responsible for adequately marking proprietary or competition sensitive information contained within their response as instructed later in this posting.

2. BACKGROUND AND SCOPE

The United States (US) Air Force Life Cycle Management Center (AFLCMC), Intelligence, Surveillance and Reconnaissance & Special Operations Forces (ISR & SOF) Directorate, Special Warfare (SW) Program Office (AFLCMC/WISN) and Contracting Office (AFLCMC/WISK) are conducting market research to identify possible sources with the ability to transmit a Multiple In-Multiple Out (MIMO) signal Mobile Ad Hoc Network (MANET) waveform in the S, and C bands with the ability to cancel interference, and to null jamming beams. The waveform should operate at or below the local noise floor and enable over 250 miles of Air-Ground Line of Sight communication. The system needs to be compatible with the AN/PRC-163 to support cross-branding data.

Currently, the USAF dismounted operators will be forced to employ at or near the forward edge to conduct operations, including, but not limited to Tactical control of airborne strike assets, Special Tactics, and Assault Zone Operations. This could involve forward staging at dispersed forward operating sites up to 250 NM from main operating bases and command and control (C2) nodes to support Combat Air Forces in combat. These forward operating sites may face enemy attempts to locate friendly force electromagnetic emissions, line of site/Over the Horizon (OTH) communications, and global positioning systems jamming. Adapting communication and jamming capabilities to counter near peer detection are essential to ensure uncontested access and the ability to freely maneuver throughout the range of military operations around the globe. This capability will allow dismounted

operators to extend their radio frequency MANET connections through denied and degraded RF environments.

3. DEFINITIONS

Key Performance Parameter (KPP): capability crucial to the operational utility or operational suitability of a solution.

Key System Attribute (KSA): capability that is important to achieve for operational utility or operational suitability.

Threshold: Threshold performance value required to be met initially by the system.

Objective: Objective performance value desired to be met by the system.

System Requirements:

Key Performance Parameters (KPPs)

1. Materiel Availability

Threshold (T) = Objective (O) Materiel availability of components shall be sufficient to maintain > 80% of the 34 8IAZT STT Global Access UTCs for 30 days of operations without failure

2. Operational Availability

T=O Each capability shall have an operational availability of > 80% over a 30-day period of 10 cycles of 72-hour missions

3. Throughput

T= Must produce >2 Mbps at 250mi

O= 10-20 Mbps at 250m

4. T=O Must operate in S, and C band

O= Shall operate in L, S, and C band

T=O Coded Orthogonal Frequency Division Multiplexing

T=Multiple In-Multiple Out signal degrees of freedom-Spatial Multiplexing

5. Low Probability of Intercept / Low Probability of Detection / AJ / Waveform threat protection

T=O LPI /LPD via power management techniques and ability to receive without transmitting

T=O AJ capabilities to detect and react to interference

T=O Capable of waveform protection

6. Antennas

T= Dual band Omni antennas with 4 dBi in S and in C band

O=MIMO compatible Tri-band Omni antennas with 4 dBi in L, S, and in C band

T= Dual band Horizontal polarized Omni antennas with 4 dBi in S and in C band

O= Tri-band Horizontal polarized Omni antennas with 4 dBi in L, S, and in C band

T= Dual band vertical polarized omni antennas with 4 dBi in S and in C band

O=Tri-band vertical polarized omni antennas with 4 dBi in L, S and in C band

T=O Directional Antenna S-Band operational frequency range of 2.2 - 2.5 GHz, and a peak gain of 12 dBi.

Vertical polarization Horizontal AZ Beamwidth 117 Degrees Vertical EL Beamwidth 18 Degrees

T=O Directional Antenna C-Band operational frequency range of 4.4 - 5.0 GHz, and peak gain of 12 dBi.

Vertical polarization Horizontal AZ Beamwidth 110 Degrees Vertical EL Beamwidth 18 Degrees

T=O 2X2 MIMO 90° Sector Antenna, Slant 1X45° L/R Polarized, 4.4 - 5.0 GHz, 16 dBi Gain
T=O 2X2 MIMO 90° Sector Antenna, Slant 1X45°L/R Polarized, 2.2 - 2.5 GHz, 15 dBi Gain

7. Interoperability

T= Will accept Ethernet and USB connections
O= Wi-Fi

8. Security

T= AES-256 encryption (FIPS 140-3)
O= Quantum resistant algorithms

9. Environments / Operating Temp

T= Must be IP67 rated
O= IP 68 Rating
T=O -40 °C to +70 °C

10. Size Weight and Power (SWaP)

T=O 5 x 3 x 2 inches
T=O Weight < 3lbs
T=O Input voltage 9-20VDC
T=O Power consumption
5 W - 16 W @ 1W TX Power
5 W - 24 W @ 4W TX Power
5 W - 48 W @ 10W TX Power

11. Kitting

T= Gear must be delivered in a 2 or 3 pelican case based kitted solution. Radios, cables, batteries, omni antennas must be delivered in one case; directional antennas and coaxial cables will be delivered in a second case. Each case must be airline checkable in size, and will not exceed 32kg per case

4. INSTRUCTION TO POTENTIAL RESPONDENTS

Interested parties who desire to participate in this SSS/RFI are encouraged to submit responses that demonstrate they presently have the technology and qualifications to provide the system described above. Responses shall also include the following information:

- a. Respondent Data:
 - Company/Institute Name
 - Address
 - Point of Contact (to include phone number and email address)
 - CAGE Code
 - Web Page URL

Pursuant to NAICS Code 334220, Radio and Television Broadcasting and Wireless Communications Equipment Manufacturing, state whether the company is:

Small Business	(Yes / No) Self-Certified or Third Party Certified
Women-Owned Small Business	(Yes / No) Self-Certified or Third Party Certified
Economically Disadvantaged Women-	

Owned Small Business	(Yes / No) Self-Certified or Third Party Certified
HUBZone Certified	(Yes / No) Self-Certified or Third Party Certified
Veteran Owned Small Business	(Yes / No) Self-Certified or Third Party Certified
Service Disabled Veteran Owned Small Business	(Yes / No) Self-Certified or Third Party Certified
8(a) Certified	(Yes / No) Self-Certified or Third Party Certified
SBA Certification Date: _____	SBA Graduation Date: _____

State whether the company is domestically or foreign-owned (if foreign, indicate the country of ownership).

State whether the respondent is interested in being a prime or subcontractor for this requirement.

- b. **Provide a rough cost estimate (i.e. rough order of magnitude (ROM)) based on an optimum or varied pricing quantities (this is for Government budget planning purposes and will not be considered a bid or binding in any manner).**
 - 1) **If development and/or integration is required to meet the system requirements, the Government also needs an estimate of the development cost based on complexity (for planning purposes only) to include PHS&T, such as warranties, portable storage, special tooling required, and training.**
- c. Describe the capability of your product or products and provide product information and specifications. Additionally, describe the capability of your product(s) to meet the desired performance requirements listed below. If you are proposing development to integrate existing or new subsystems, clearly describe your approach.
- d. Responses may include but are not limited to:
 - 1) Pictures
 - 2) Training videos
 - 3) Architecture views
 - 4) Drawings
 - 5) Associations or connectivity with other end items/systems
 - 6) Test results to support meeting threshold and objective requirements
 - 7) Ability to deliver within 12 to 18 months after receipt of order
- e. Indicate whether the company is currently performing this type of work/whether the capability can be obtained under an existing Government contract. Provide the contract number(s) and Government points of contact accordingly.
- f. Indicate whether the company currently provides this type of work as a commercial solution. In this instance, “commercial item” means an item that is of a type customarily used by the general public or non-Governmental entities for purposes other than Governmental purposes. Please indicate if this item has been sold, leased, or licensed to the general public; or has been offered for sale, lease, or license to the general public. Indicate whether all materials used are in compliance with the *Buy American Act* or *Berry Amendment* (10 U.S.C. 2533a), as applicable.
- g. Please include product warranty terms as applicable, along with any warranty limitations. If product warranty terms are not applicable, please include N/A.

- h. Include any spare parts that would potentially be included.
- i. Provide the details of NET, training, including process and cost.
- j. Describe any possible product improvements that could be made or are planned for the future for the solution.
- k. Describe any risks associated with the fulfillment of this projected requirement. This can include technical or schedule risks associated with development, key performance parameters, etc.
- l. Please provide specific details and rationale as to how compliance with FAR 52.219-14, *Limitations on Subcontracting*, and FAR 52.219-33, *Non-Manufacturer Rule*, would be achieved in the event the Government sets this requirement aside for small business. Include specific details regarding teaming arrangements, etc. If subcontracts are to be used, provide the anticipated percentage of effort to be subcontracted and whether small or large businesses, including foreign, will be used. If this effort is not set-aside for small business, small business utilization will be considered. Large and small businesses should provide a reasonable expectation for small business utilization as a percent of total contract value as well as supporting rationale for the recommended percentage.

5. RESPONSES, COMMUNICATIONS, AND ADDITIONAL CONSIDERATIONS:

- a. The Government may contact respondents to obtain additional information or clarification in order to fully understand a company's response. This may include discussions, site visits, demonstrations, etc. to further the Government's understanding of a proposed solution, as well as a company's understanding of the Government's requirements.
- b. Responses shall not exceed 15 pages (including cover page). Include technical and descriptive text and photographs/illustrations to provide sufficient detail. The submitted documentation becomes the property of the United States Government and will not be returned.
- c. Respondents shall clearly mark proprietary data with the appropriate markings. Any proprietary information received in response to this request will be properly protected from any unauthorized disclosure. Any material that is not marked will be considered publicly-releasable.
- d. Government-contracted contractor support personnel may handle proprietary data submitted in response to this RFI. These support personnel will have signed the same Non-Disclosure Agreements (NDAs) as Government personnel.
- e. **If you are currently providing your proposed capability on an existing government contract, please identify the contract and a government POC.**
- f. Broad industry participation for this RFI is encouraged. U.S. and non-U.S. firms are permitted to respond.
- g. Please note that prospective contractors must be registered in the System for Award Management (SAM) database to be awarded a DOD contract; the NAICS code for this requirement, if applicable to the company's capabilities, should be included in the company's NAICS code

listing in SAM. To learn more about this SAM requirement and how to register, please visit the SAM website at: **<https://www.sam.gov/portal/SAM>**

- h. Responses are due electronically by 12:00PM Eastern time on 26 March 2026 via email to:

Jamie Sclafani
AFLCMC/WISK
Contracting Officer
jamie.sclafani@us.af.mil

Cesar Palacios-Maya
AFLCMC/WISN
Program Manager
cesar.palacios-maya.1.ctr@us.af.mil

- i. This notice may be updated as additional information becomes available. Questions regarding this announcement shall be submitted in writing by email to the Contracting Officer identified above, with the other individuals identified above copied as a courtesy.

This is a Request for Information (RFI) only to identify potential sources for the solution. The information provided in the RFI is subject to change and is not binding on the Government. The Government has not made a commitment to procure any of the items discussed, and release of this RFI should not be construed as creating such a commitment or as an authorization to incur cost for which reimbursement would be required or sought.